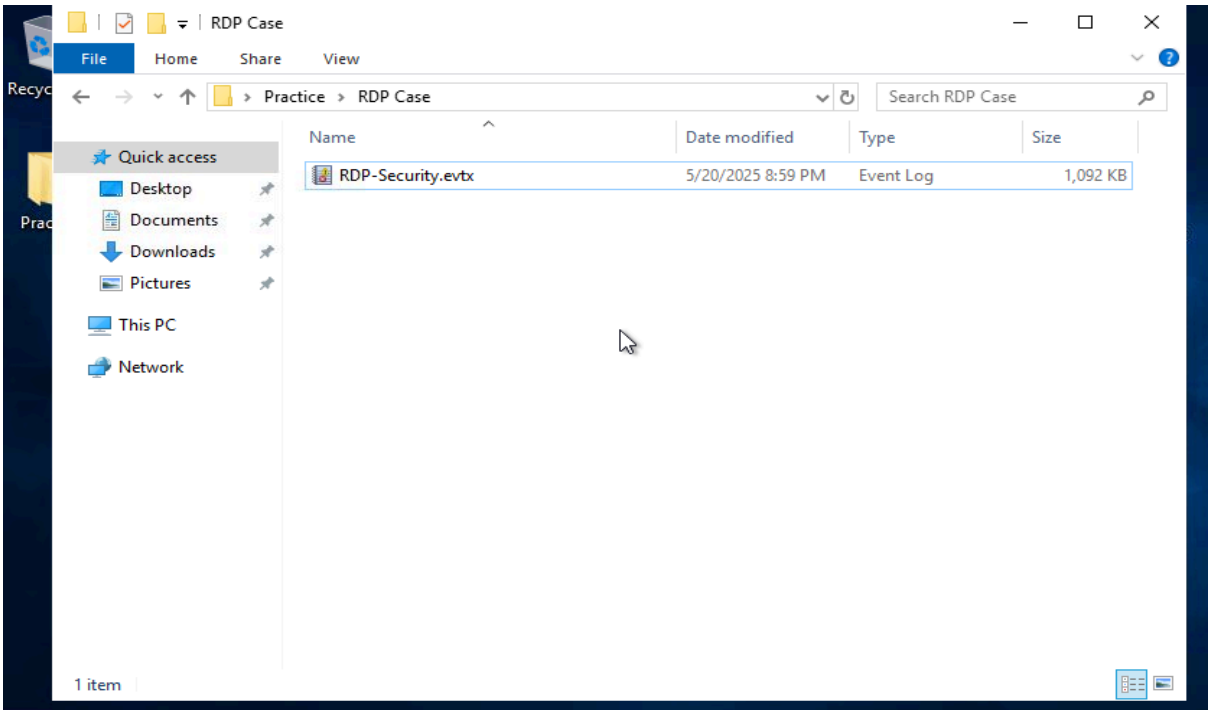
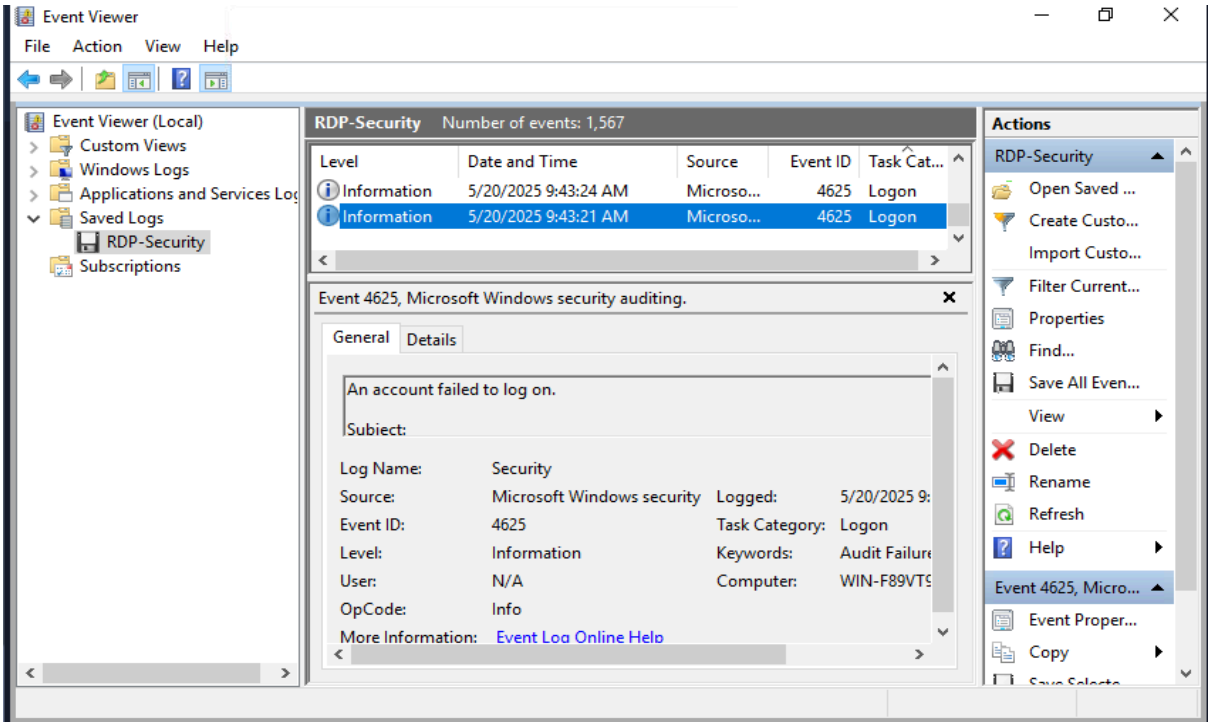
Basic threat detection and windows log analysis

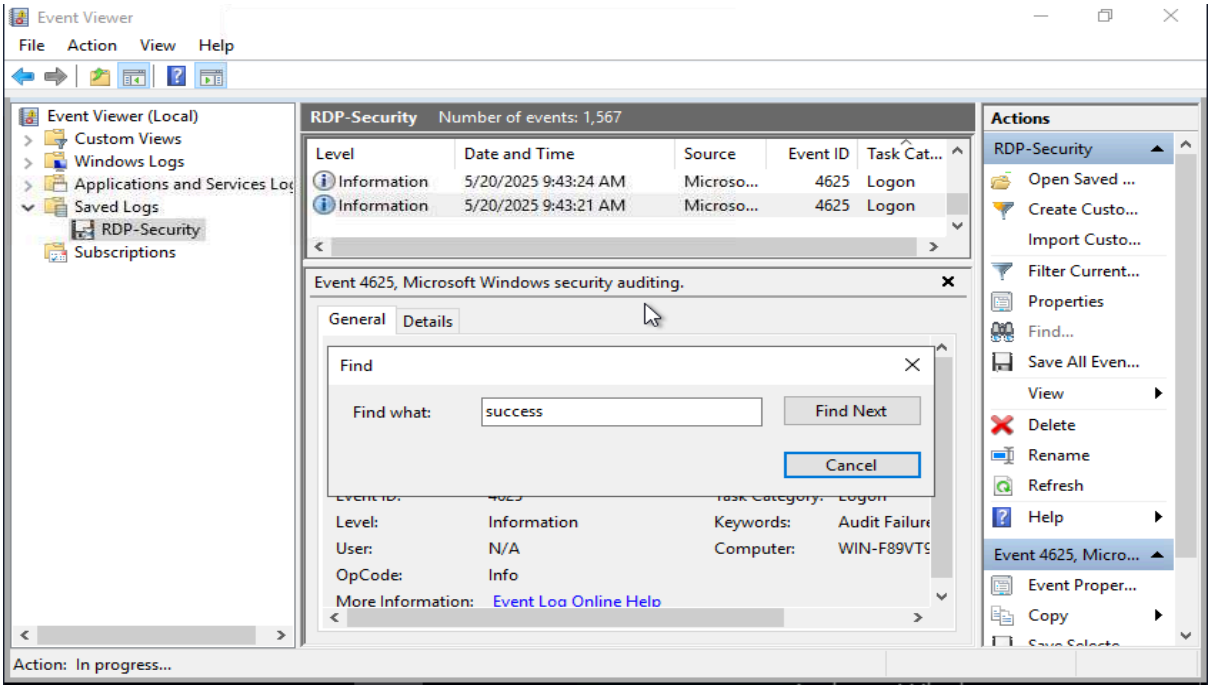
In this virtual lab we are tasked to investigate a brute force attack by analyzing windows logs , RDP-Security.evtx is the file we'll need to examine .

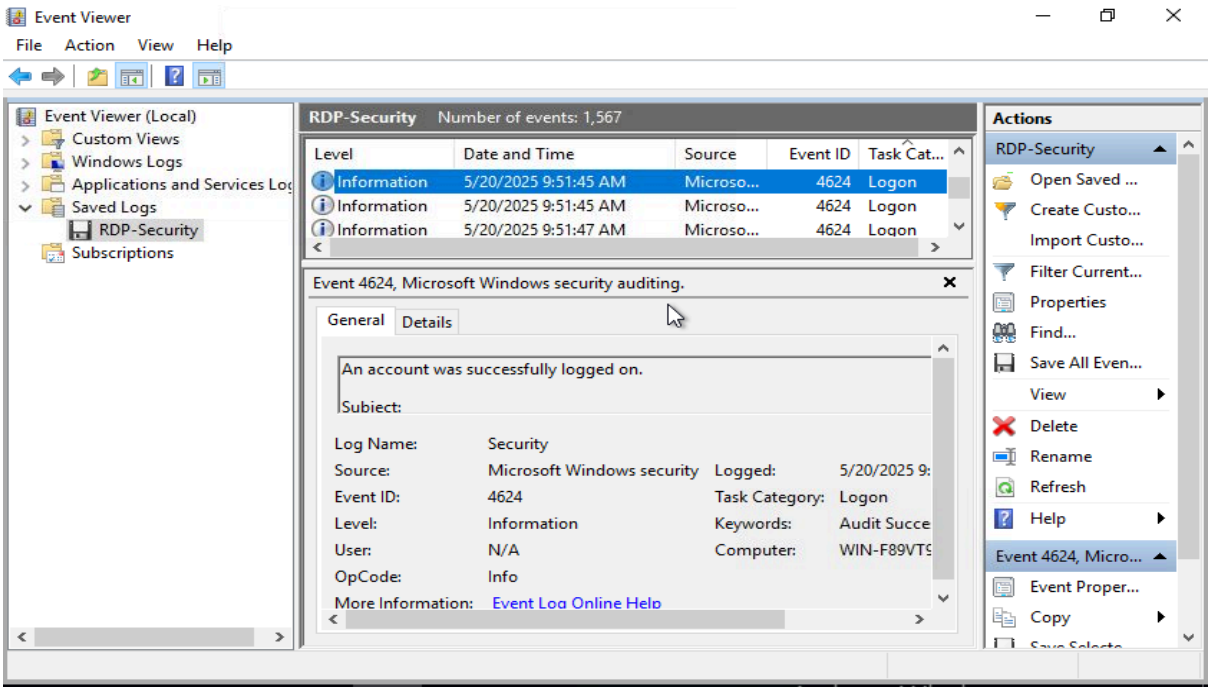




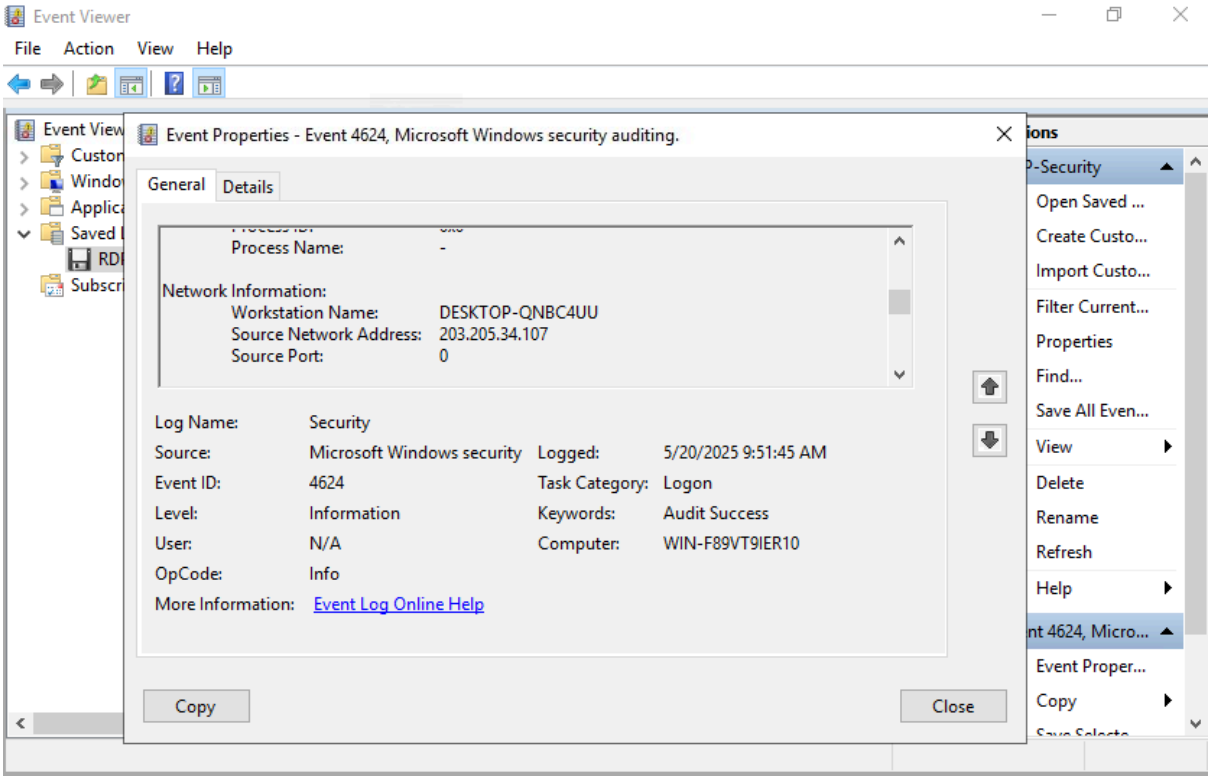
After checking the saved logs file , we can notice many failed

attempts of logging in to the administrator account . The rule of this event (ID 4625) is that 3 failed attempts need to occur .
About 1567 logs of the same event were generated , caused by multiple IP addresses which indicates a botnet type of brute force attack targeting the administrator account .
First I need to check how many of the events were caused by the attacker and if there are any successful logins .
I can do that by filtering and checking for the word “success”





The first successful attempt (Event ID 4624) occurred at 9:51:45 AM by ip address 203.205.34.107 , Workstation name : DESKTOP-QNBC4UU



The password was changed by the attacker after their successful login , the duration of this attack was 41 minutes and 44 seconds .

This unfortunate accident could have been avoided by applying a strong password (15 random letters digits and chars) a minimum and regularly changing it in order to defend against dictionary attacks , adding multi factor authentication and a lockout policy after many unsuccessful attempts and applying strict firewall rules.

The appropriate response should be to isolate the compromised system , reset all the passwords and continue a full log inspection to check for the attacker’s activity and persistence techniques , as well as performing a full malware analysis and escalating the incident to higher level operators in order to consider our next procedures .